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Traffic Signal Priority Control Strategy for Connected Emergency Vehicles with Dilemma Zone Protection for Freight Vehicles --Manuscript Draft--

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Traffic Signal Priority Control Strategy for Connected Emergency Vehicles with Dilemma Zone Protection for Freight Vehicles

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This paper presents an emergency vehicle priority control systems based on Connected Vehicle technology, called MMITSS Priority. Traditional preemption does not consider the current traffic situation, such as the presence of a freight vehicle in the dilemma zone on an opposing movement, and can have a significant negative impact on the minor movement vehicles. A mixed-integer linear programming model is developed which can consider the priority requests from multiple emergency vehicles and dilemma zone requests from freight vehicles that could be trapped in the dilemma zone. The optimization model provides an optimal schedule that minimizes the total weighted priority request delays and dilemma zone request, as well as some flexibility to adapt to other vehicles in real time. The flexible implementation of the optimal signal timing schedule is designed to improve the mobility of the non-emergency vehicles. The approach has been tested and evaluated using microscopic traffic simulation. The simulation experiments show that the proposed priority control method is able to improve the travel time of the vehicles on the minor street while ensuring safe passage of the freight vehicle at the dilemma zone without significantly delaying the emergency vehicles. The method will be implemented in the Maricopa County SMartDRIVE ProgramSM test bed in Anthem, Arizona.

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1 INTRODUCTION

2 Traffic signal control is responsible for regulating traffic flow in a signalized intersection
3 and for improving traffic mobility and safety. Traffic signal control can also be operated to provide
4 right-of-way priority to emergency vehicles such as ambulances, fire trucks, police cars, etc.
5 According to the Manual on Uniform Traffic Control Devices (1), a traffic signal controller can
6 alter the regular signal timing and phasing to provide right-of-way for emergency vehicles. The
7 alternative signal timing and phasing may either extend the currently displayed green interval or
8 replace the entire set of signal phases and timing depending on the emergency vehicle's requested
9 signal phase (approach). Historically, the emergency vehicle sends a preemption request to the
10 traffic signal controller by using either optical, acoustic, special inductive loop technology, or
11 Global Positioning System technology (2). Traffic signal cabinets are equipped with an emergency
12 vehicle preemption device to receive the preemption request. In 1970, Long (3) developed a
13 preemption system where emergency vehicles transmit an optical signal and the receiver detects
14 the presence of the emergency vehicles. The detector then commands the signal controller to flash
15 a green light for the requested phases.

16 Emergency vehicles need to respond to the emergency incidents as quickly as possible.
17 Statistics have shown that slow response to emergency calls increases the casualty rate of the
18 victims. Fogue (4) and Li (5) cited that 10% of casualties occur within just a few minutes or
19 seconds after accidents. The survival rate of a patient is reduced by 7-10% with every elapsed
20 minute (6). Emergency vehicles experience delays at intersections due to long queues and when
21 stopping at the traffic signal to ensure a safe right-of-way. To improve the mobility of the
22 emergency vehicles, especially for fire response vehicles, an emergency vehicle signal preemption
23 (EVSP) systems was developed by Gordon et al. (7). The EVSP system provides a green wave to
24 emergency vehicles when the first responders manually push a button in the fire station as they
25 depart. The green wave remains active along the response route until the fire vehicle passes each
26 intersections. The traffic signal returns to its normal state after a fixed amount time. Qin et al. (8)
27 modified EVSP to implement a real-time control strategy instead of fixed time phase holds. The
28 real-time control strategy detects the presence of the emergency vehicles using connected vehicle
29 technology and allocates the maximum green time required for the emergency vehicle to pass the
30 intersections. Shaaban et al. (9) proposed an effective signal preemption and path selection
31 strategy. The optimal path is selected from the starting location of the emergency vehicles to the
32 destination of the emergency call. This system uses vehicle-to-vehicle (v2v) and vehicle-to-
33 infrastructure (v2i) technology to notify other vehicles about the emergency vehicle information
34 which may help to clear any traffic queue before the emergency vehicle arrives at the intersection.
35 Chen et al. (10) claim that the shortest path cannot ensure earliest arrival time. The emergency
36 vehicles can respond to the emergency call more quickly by travelling through the path with better
37 traffic conditions. They developed an emergency vehicle path selection algorithm which considers
38 current and historical traffic information. Emergency Vehicle Response Time (EVRT) is another
39 traffic signal control method for emergency vehicles preemption. The EVRT strategy utilizes
40 connected vehicle technology to predict the queue length and provide an early green to clear the
41 queue at the downstream intersections for improving the mobility of the emergency vehicles (11).
42 To predict the queue length, EVRT method considers 100% penetration of connected vehicles.

Traffic signal preemption control generally serves only one, or possibly multiple emergency vehicles that are traveling on the same approach to an intersection. When multiple emergency vehicles arrive from different approaches simultaneously, currently traffic signal control generally follows first-come-first serve rules (12), but there are mechanisms where one approach (preemptor) can override others. There have been accidents where two emergency vehicles have collided in an intersection (13,14). Gross et al. (15) proposed a system which provides early warning of the arrival and departure of emergency vehicles along with the direction from which the emergency vehicle is approaching. The emergency vehicle uses a transceiver to transmit its travel direction. The transceiver also receives the travel direction messages of other emergency vehicles which are within range of the transceiver. This system allows the emergency vehicles drivers to determine the possibility of collisions with other emergency vehicles. Goel et al. (16) developed an intelligent traffic signal system where a wireless sensor network (WSN) has been employed to communicate between the intersections. The WSN detects the emergency vehicle based on its emitted sounds. The sensor sends the information to next intersection over the WSN. If multiple emergency vehicles are approaching from different directions, the intersection serve them based on a first-come-first serve rules.

It is a significant challenge for the traffic engineers to ensure safe passage of the emergency vehicles. Researchers at the University of Washington found that from 1994-2002, approximately 300 fatalities occurred annually in the U.S. during police pursuits (17). According to Fahy (18) and Bui et al. (19), approximately 94 fatalities resulted from the collisions between a fire service vehicle and other vehicles, pedestrians or bicyclists from 1997-2006. The National Highway Traffic Safety Administration (NHTSA) released a report in April 2014 based on analyzing the number of ambulance accidents occur from 1992-2011 (17). The report found that ambulances are involved in approximately 6,500 accidents each year. The report also found that nearly 60% of ambulance accidents occur while the vehicles are in emergency use. Emergency vehicles rely on vehicle horn, sirens, and flashing lights to alert the vehicles and pedestrians to prevent accidental collisions with pedestrians or other vehicles (20). Excessive use of lights and sirens during emergency driving may cause accidents (21,22). The light and sirens can limit the auditory capability of the emergency vehicle drivers and impede the vision of adjacent road users. Smith et al. (23) developed a warning system which provides indication to the vehicles and pedestrians about the approaching direction of the emergency vehicles. Buchenscheit et al. (24) use v2v and v2i communication systems to notify the nearby vehicles about the presence, location and speed of emergency vehicles. Bui et al. (19) suggested that comprehensive emergency vehicle driver training and risk management may be effective approaches to reduce the emergency service vehicle incidents.

Traffic signal preemption cannot directly address some of the concerns associated with safety, but, if done intelligently, can reduce delay and provide less disruption to the logical traffic signal control at intersections and along response routes, which should improve safety. Current traffic signal preemption generally overrides other traffic signal control strategies, such as coordination, to force-off the current phases to immediately display green signal indications ahead

of the emergency vehicles. One potential scenario occurs when other vehicles, such as freight trucks or other heavy vehicles are also approaching a preempted intersection. The operating characteristics of freight trucks are significantly different from the passenger cars. Heavy vehicle deceleration depends on the mass, tire-pavement friction, braking efficiency, and reaction time of the drivers. Gates et al. (25) found that due to the low deceleration capability, heavy vehicles are 3.6 times more likely to be red light runners compared to passenger vehicles. Accidents may occur when the traffic signal controller attempts to terminate the current signal phase without considering the presence of vehicles in the dilemma zone. Tarko et al. (26) developed a probabilistic algorithm which identifies the optimal green extension for the vehicles in the dilemma zone. McCoy et al. (27) developed an advanced detection and warning system to improve the level of dilemma-zone protection. Zimmerman et al. (28) designed a detection-control system (D-CS) which determines the phase termination time based on the number of vehicles in the dilemma zone, and the waiting time of vehicles on conflicting phases. None of these systems consider the dilemma-zone during a preemption event.

Emergency vehicle preemption systems disrupt the normal traffic operations to provide right-of-way to the emergency vehicles. Studies have been conducted to analyze the impact of the traffic signal preemption on the traffic flow. Bullock et al. (29), Nelson et al. (30), and Collura et al. (31) found that preemption system can negatively impact arterial travel time of the responding vehicles. Paruchuri (32) developed an adaptive preemption of traffic (APT) system which considers real time traffic data to reduce the vehicle delay of the non-emergency vehicles.

Traffic signal preemption systems aim to reduce travel time for emergency vehicles, but the operations and benefits are not always clear. When a vehicle requests preemption, the quick change in control priorities can significantly disrupt traffic flow and may create confusion that could impact safety. The current approach to preemption makes control decision that are independent of the current traffic situation. For example, consider a fire response vehicle traveling along an arterial where the signal controllers are changing to create a green pathway for the vehicle. If there are other emergency vehicles on other approaches or perhaps a freight vehicle (e.g. a gasoline tanker truck) in the dilemma zone, they may be caught unaware and may not be able to safely stop. This paper presents a Connected Emergency Vehicle (CEV) application which ensures smooth progression of single or multiple emergency vehicles through the corridors, allows safe passage of the heavy vehicles which are in the dilemma zone, and improves the overall performance of the traffic signals. The emergency vehicle signal priority framework presented in this paper is based on the priority model described in (13) which is implemented in the Multimodal Intelligent Traffic Signal System (MMITSS) project. MMITSS is a U.S. Department of Transportation Cooperative Transportation Systems Pooled Fund project that focuses on connected vehicle traffic signal applications based on v2v and v2i wireless communication technologies.

SYSTEM ARCHITECTURE

The Connected Emergency Vehicles preemption system architecture (Figure 1) consists of six hardware devices: Roadside unit (RSU), signal controller, MMITSS Roadside Processor (MRP), onboard unit (OBU), vehicle-side processor (VSP), and a display unit for the human machine interface (HMI) (which is not required for deployment, but useful for testing and demonstrations). The RSU and OBU are responsible for wireless message transmission and currently use DSRC but could use a variety of wireless technologies including C-V2X or other 5G NR technologies. The software components reside on the MRP and VSP. The HMI is an in-vehicle display unit that runs on a laptop or other computer. It can provide information about the current signal status, priority status, and presence of other connected vehicles. The HMI can be integrated with any vehicle's existing HMI system through a JavaScript Object Notation (JSON) formatted application programming interface (API) message, sent from the VSP.

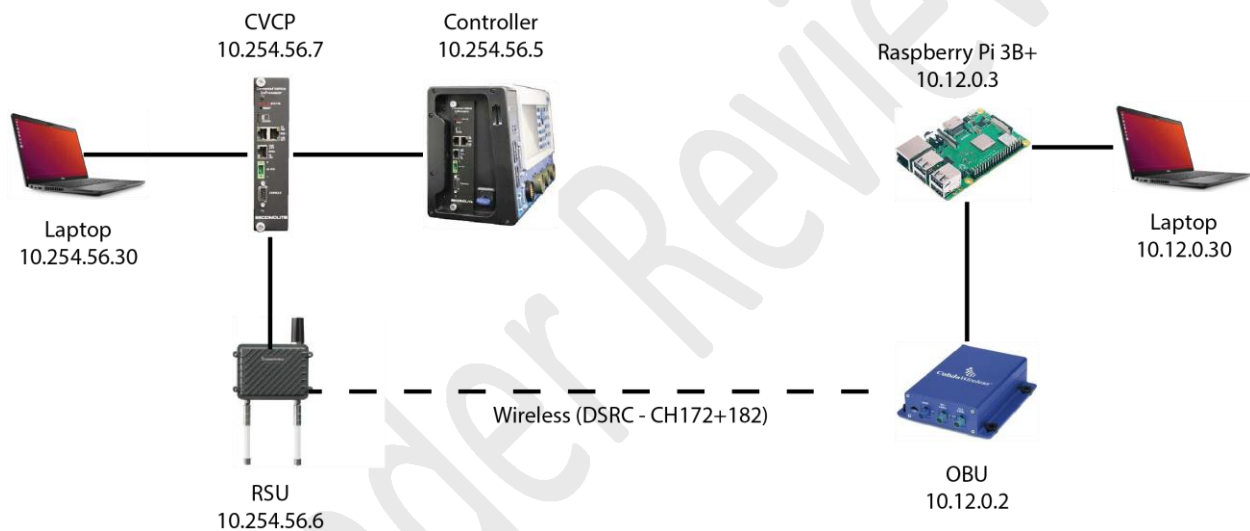


Figure 1: MMITSS System Architecture

METHODOLOGY

In the MMITSS signal priority control system, the MAP message is broadcasted by an RSU. The MAP message is developed by using “USDOT connected vehicles tools-ISD Message Creator” (33). The MAP message resides on the MRP. All the received MAPs are stored in a ‘Received Map’ folder inside the VSP. A “Map Engine Library” is used to locate a vehicle on one of the available maps based on the vehicles current GPS coordinates and map reference points. After obtaining an active map, the VSP calculates the vehicles expected arrival time (ETA) at the intersection. The VSP then formulates a signal request message (SRM) in SAE J2735 2016 standard format which is broadcasted by OBU.

The RSU receives the SRMs from different priority eligible vehicles and forwards them to the software component (message transceiver) which reside on the MRP. The priority request server (PRS) is a software component that manages a list of active priority requests and formulates signal status message (SSM) as an acknowledgement message for the connected vehicles. Each

active vehicle receives the broadcasted SSMs and validates that the request that was sent is accurate and then forward the SSM information to the HMI for display. The PRS forwards the priority requests list to the Priority Request Solver (PRSolver) software component.

Figure 2 shows the control logic of the PRSolver. The PRSolver receives the current signal plan and current signal status from the Traffic Controller Interface (TCI) to formulate a mathematical optimization model. The PRSolver checks the vehicle type in the priority request list to determine if there is any priority request from emergency vehicles. If there is, other requests, such as transit priority requests, are denied, leaving only the active emergency vehicle requests. Then, the preemption signal priority logic checks to see if there are any freight vehicles in the dilemma zone. If there are, freight vehicle dilemma zone requests are added to the optimization model.

Based on the active priority requests, the current signal plan and the signal status, the preemption signal priority algorithm identifies the required signal group and formulates the mathematical optimization model. A GNU GLPK package (version 4.55) is used to solve the optimization model. A list of optimal signal control events (schedule) is generated based on the optimal solution. The event list is sent to the TCI as a signal timing schedule. The TCI is responsible for executing the signal schedule using NTCIP commands.

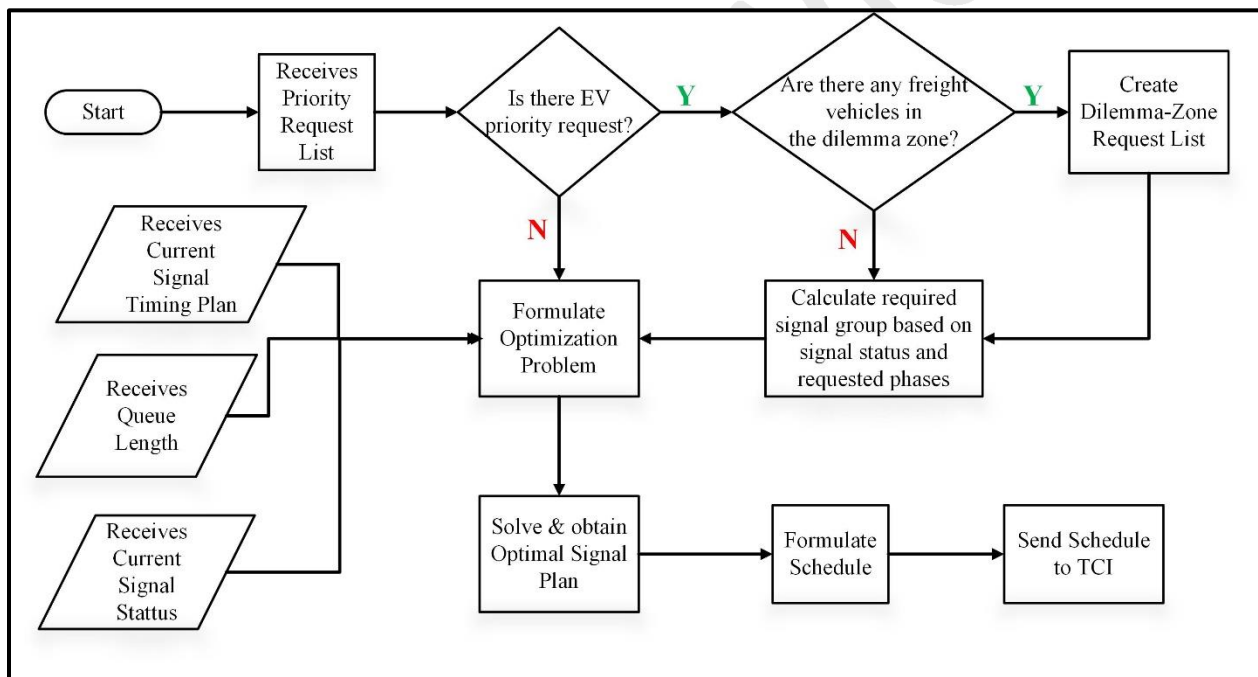


Figure 2: Logic Control of Priority Request Solver

Calculation of Dilemma Zone Request

When a traffic signal changes from green to yellow, vehicles approaching the intersection have to decide if they can safely stop or pass the intersection. When there is some uncertainty about this decision, the vehicles are defined as being in the dilemma zone (25). Gates et al. (34) found that heavy vehicles are less likely to stop if they are in dilemma zone. The algorithm presented in this paper considers the presence of connected heavy vehicles in the dilemma zone as a dilemma zone priority request. The algorithm calculates the stopping sight distance (SSD) to

estimate whether a heavy vehicle is in dilemma zone or not. Stopping sight distance is a function of vehicle speed, reaction time of drivers, coefficient of friction between tires and roadway and roadway grade (35-37). The SSD equation recommended by the AASHTO Greenbook is following:

$$SSD = 1.47Vt + \frac{V^2}{30(f \pm G)} \quad (1)$$

where,

V= design speed (mph)

t = perception-reaction time (sec)

f = coefficient of tire-pavement friction, and

G = roadway grade.

The stopping sight distance would be different for heavy vehicles, passenger vehicles, motorcycles, and other vehicles based on the different vehicle characteristics.

Estimating Queue Length

There are several approaches that can be used to estimate queue length. With the advent of Connected Vehicle technology it is possible to estimate the queue length of an approach in advance (38). Tiaprasert et al. (39) collected basic safety messages (BSMs) to estimate travel time and queue length using a v2i communication systems. Liu et al. (40) use approach and stop bar detectors to estimate key shockwave points to estimate queue length. When the market penetration of connected vehicles is low, a combination of these approaches is needed. Koonce et al. (41) cited that queue clearance time can be calculated by approximating the startup loss time as two seconds and the required green time to clear the intersection as two seconds per vehicle in the queue. The signal priority control strategy proposed in this paper, uses this approximation and connected vehicle technology to estimate the queue clearance time and queue length, respectively.

MATHEMATICAL MODEL

The traffic signal control algorithm for connected emergency vehicles is developed based on the design, framework and algorithm of the MMITSS project (13,42). The optimization model can accommodate multiple, multi-modal priority requests with explicit consideration of the priority requests from the heavy vehicles in the dilemma zone. The optimization model is formulated as a Mixed Integer Linear Programming (MILP) problem to minimize the priority delay and the negative impact on the regular vehicles. To minimize the negative impact on the regular vehicle, a flexible implementation algorithm is developed. The flexible implementation algorithm allows the non-priority phases to be actuated. Therefore, it is possible to serve dynamic traffic demand in real-time.

Model Requirements

- The model will not shorten or omit the yellow change interval, or any red clearance interval (1,40).
- The optimal signal timing schedule will not permit a signal indication sequence to transition from a steady yellow signal indication to a steady green signal indication (1,40).

The mathematical model notation is summarized in Table 1.

1 **Table 1:** Notation definition of Sets, Indices, Parameters and Variables

Types	Symbol	Definition
Sets	K	Set of Cycles
	P	Set of Phases
	T	Set of All modes
	J	Set of number of priority request
	E	Set of critical points for priority request
	Z	Set of number dilemma zone request
	$P11$	Set of phases in ring 1 barrier group 1
	$P12$	Set of phases in ring 1 barrier group 2
	$P21$	Set of phases in ring 2 barrier group 1
	$P22$	Set of phases in ring 2 barrier group 2
Indices	$k \in K$	Index of cycles
	$p \in P$	Index of phases
	$t \in T$	Index of vehicle modes
	$j \in J$	Index of priority request
	$z \in Z$	Index of dilemma zone request
	$e \in E$	Index of Critical points
Parameters	y_p	Yellow change interval for phase p
	r_p	Red clearance time for phase p
	$gmin_p$	Minimum green time for phase p
	$gmax_p$	Maximum green time for phase p
	$SP1$	Starting phase of ring 1
	$SP2$	Starting phase of ring 2
	$elapsed1$	Elapsed green time of $SP1$
	$elapsed2$	Elapsed green time of $SP2$
	$init1$	Required time to start next phase in ring1 if the current phase is yellow change or red clearance
	$init2$	Required time to start next phase in ring2 if the current phase is yellow change or red clearance
	w^t	Priority weight for different modes of vehicle
	w^z	Weight for the vehicles in dilemma zone
	q_p	Queue clearance time per phase
	Rl_p^j	Earliest arrival time of the j^{th} priority request of mode t at the stop bar
	Ru_p^j	Latest arrival time of the j^{th} priority request of mode t at the stop bar
	Dl_p^j	Earliest arrival time of the z^{th} dilemma zone vehicle at the stop bar
	Du_p^j	Latest arrival time of the z^{th} dilemma zone vehicle at the stop bar
	a_p^j	Binary data, $\{1, \text{for the requested signal phase by } j^{th} \text{ priority request}$ 0 otherwise
	b_p^j	Binary data, $\{1, \text{for the requested signal phase by } z^{th} \text{ dilemma zone request}$ 0 otherwise

Variables	M	A big positive number
	α	A small positive fractional number
	I_p^k	An Indicator $\begin{cases} 0 & \text{if phases are already served in cycle 1} \\ 1 & \text{otherwise} \end{cases}$
	$Grn1_p^k$	Passed green time for all phases in cycle 1, ring1 $\begin{cases} 1 & \text{if } p = SP1 \text{ and } k = 1 \\ 0 & \text{otherwise} \end{cases}$
	$Grn2_p^k$	Passed green time for all phases in cycle 1, ring2 $\begin{cases} 1 & \text{if } p = SP1 \text{ and } k = 1 \\ 0 & \text{otherwise} \end{cases}$
	$gmax_Ring_p^k$	Maximum green time Per Ring, $gmax_p[p]$
	$t_{p,e}^k$	Starting time of phase p in cycle k for critical points e
	$g_{p,e}^k$	Green time for phase p in cycle k for critical points e
	v_{pe}^k	Duration of phase p in cycle k for critical points e , including clearance time
	d_p^j	Delay of j^{th} priority request
	γ_p^z	Delay of z^{th} dilemma zone request
	θ_p^j	Binary Variable, $\begin{cases} 0, & \text{if the } j^{th} \text{ priority request is served in Cycle 1} \\ 1, & \text{otherwise} \end{cases}$
	δ_p^z	Binary Variable, $\begin{cases} 0, & \text{if the } z^{th} \text{ dilemma zone request is served in Cycle 1} \\ 1, & \text{otherwise} \end{cases}$
	$PriorityDelay$	Total priority request delay
	$DilemmaZoneDelay$	Total dilemma zone request delay
	$Flex$	Flexibility region of signal plan
	fp^{mn}	First phase in ring m and barrier group n
	lp^{mn}	Last phase in ring m and barrier group n
	np^{mn}	Number of phase in ring m and barrier group n
	χ_p^j	Intermediate variable for the j^{th} priority request
	β_p^j	Intermediate variable for the z^{th} dilemma zone request

1 The NEMA standard dual-ring, eight phase structure is used as the basis for the
2 mathematical model formulation. Figure 3(a) displays a typical North-American dual-ring, eight-
3 phase controller. Although the proposed model is formulated based on the eight-phase dual-ring
4 structure, it can also be applied to the different phase structure such as, T-intersection. In a dual-
5 ring controller, each ring contains four phase and the barriers are used to separate the conflicting
6 movements. The barriers are also used to determine a termination point at which phases from both
7 rings must cross the barrier simultaneously (40). The precedence diagram for dual ring eight phase
8 controller is depicted in Figure 3(b). Each phase is represented as “activity on arc” in the
9 precedence diagram. The arc represents the phase duration and the node represents the phase
10 transition. The phase duration is (v_p^k) consists of green time (g_p^k), yellow change (y_p^k) and red
11 clearance time (r_p^k) ($v_p^k = g_p^k + y_p^k + r_p^k$). In an actuated signal controller, the phase duration is,
12 $v_p^k = g_p^k + e_p^k + y_p^k + r_p^k$ where g_p^k denotes minimum green time and e_p^k denotes phase extension

time. The two-step green time for a phase allows for implementation of a flexible signal timing that can extend or gap out phases based on vehicle detection. The optimal signal timing developed by the model has two critical points (left and right critical points denoted by CP_l , CP_r respectively) for each phase. The left critical point (CP_l) denotes the minimum green time or phase hold time. After the left critical point, actuated signal control logic will be implemented based on real time traffic demand as sensed by detectors. The signal phase is forced-off when it reaches the right critical point (CP_r).

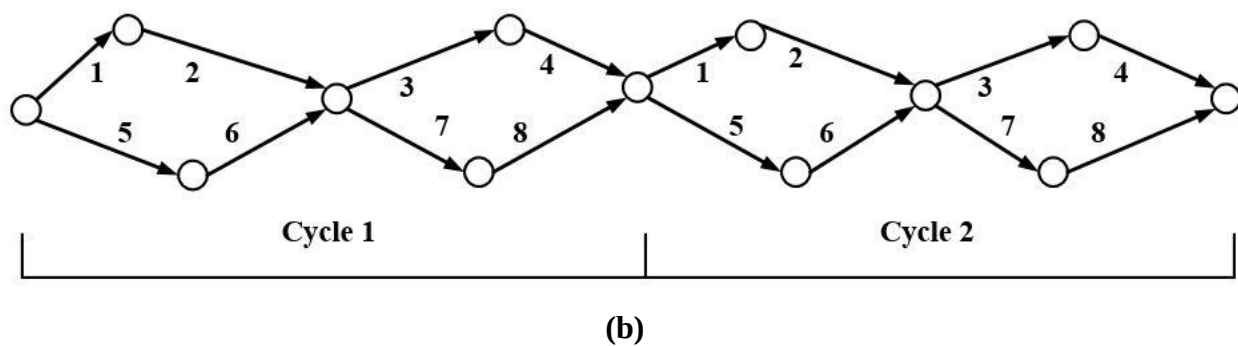
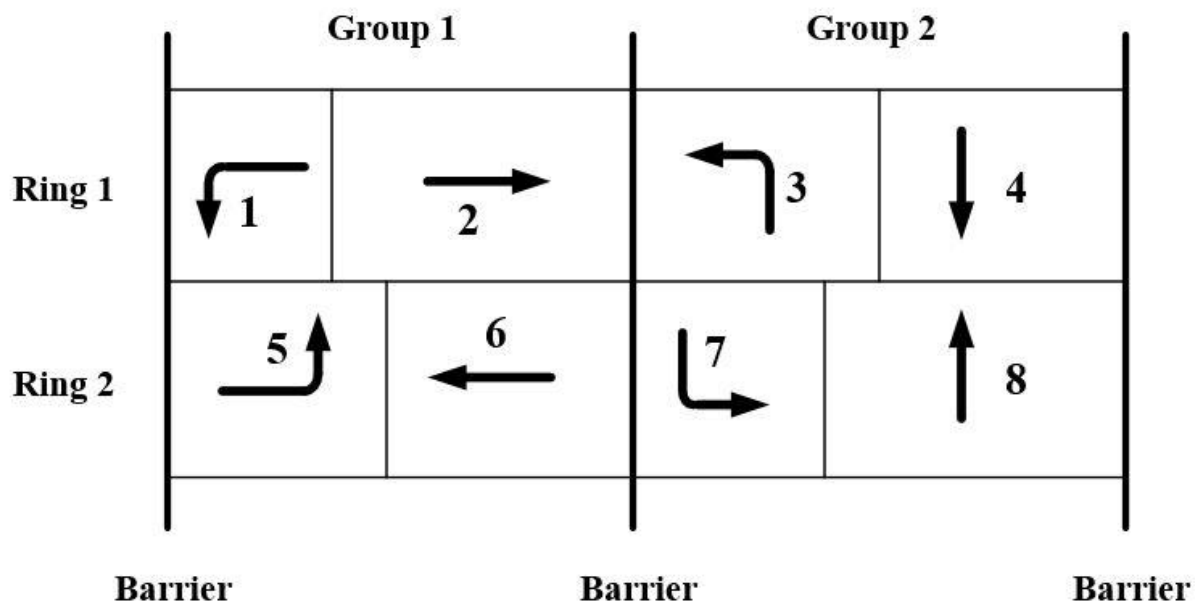


Figure 3: (a) Dual-ring, eight-phase controller (b) Precedence graph representation of a dual-ring eight-phase controller

The objective of the mathematical model is to minimize the total weighted delay of the emergency vehicles, and the heavy vehicles that trapped in the dilemma zone and maximize the flexibility of the signal timing plan. The main structure of the mixed integer linear programming model is as follows:

$$\begin{aligned} & \text{minimize} \sum_{t \in T, j \in J, p \in P} \text{PriorityDelay} + \sum_{z \in Z, p \in P} \text{DilemmaZoneDelay} - \sum_{p \in P, k \in K} \text{Flex} \\ & = \text{minimize} \sum_{t \in T, j \in J, p \in P} w^j a_p^t d_p^j + \sum_{z \in Z, p \in P} w^z b_p^z \gamma_p^z - \alpha \sum_{p \in P, k \in K} \text{Flex} \end{aligned}$$

Subject to:

Precedence Constraints:

$$t_{p+1,e}^1 = t_{p,e}^1 + v_{p,e}^1 \quad \forall e, k, p, (p+1) \in P11, p \geq SP1 \quad (2)$$

$$t_{p+1,e}^1 = t_{p,e}^1 + v_{p,e}^1 \quad \forall e, k, p, (p+1) \in P12, p \geq SP1 \quad (3)$$

$$t_{p+1,e}^1 = t_{p,e}^1 + v_{p,e}^1 \quad \forall e, k, p, (p+1) \in P21, p \geq SP2 \quad (4)$$

$$t_{p+1,e}^1 = t_{p,e}^1 + v_{p,e}^1 \quad \forall e, k, p, (p+1) \in P22, p \geq SP2 \quad (5)$$

$$t_{p,e}^k = t_{lp^{11},e}^k + v_{lp^{11},e}^k \quad \forall e, k = 1, p \in P12, np^{12} + p \leq 5, p > SP1 \quad (6)$$

$$t_{p,e}^k = t_{lp^{11},e}^k + v_{lp^{11},e}^k \quad \forall e, k = 1, p \in P22, np^{22} + p \leq 9, p > SP2 \quad (7)$$

$$t_{p,e}^k = t_{lp^{21},e}^k + v_{lp^{21},e}^k \quad \forall e, k = 1, p \in P12, np^{12} + p \leq 5, p > SP1 \quad (8)$$

$$t_{p,e}^k = t_{lp^{21},e}^k + v_{lp^{21},e}^k \quad \forall e, k = 1, p \in P22, np^{22} + p \leq 9, p > SP1 \quad (9)$$

$$t_{p+1,e}^k = t_{p,e}^k + v_{p,e}^k \quad \forall e, p, (p+1) \in P11, k > 1 \quad (10)$$

$$t_{p+1,e}^k = t_{p,e}^k + v_{p,e}^k \quad \forall e, p, (p+1) \in P12, k > 1 \quad (11)$$

$$t_{p+1,e}^k = t_{p,e}^k + v_{p,e}^k \quad \forall e, p, (p+1) \in P21, k > 1 \quad (12)$$

$$t_{p+1,e}^k = t_{p,e}^k + v_{p,e}^k \quad \forall e, p, (p+1) \in P22, k > 1 \quad (13)$$

$$t_{p,e}^k = t_{lp^{11},e}^k + v_{lp^{11},e}^k \quad \forall e, p \in P12, k > 1, I_p^k = 1, np^{12} + p \leq np^{12} + fp^{12} \quad (14)$$

$$t_{p,e}^k = t_{lp^{11},e}^k + v_{lp^{11},e}^k \quad \forall e, p \in P22, k > 1, I_p^k = 1, np^{22} + p \leq np^{22} + fp^{22} \quad (15)$$

$$t_{p,e}^k = t_{lp^{21},e}^k + v_{lp^{21},e}^k \quad \forall e, p \in P12, k > 1, I_p^k = 1, np^{12} + p \leq np^{12} + fp^{12} \quad (16)$$

$$t_{p,e}^k = t_{lp^{21},e}^k + v_{lp^{21},e}^k \quad \forall e, p \in P22, k > 1, I_p^k = 1, np^{22} + p \leq np^{22} + fp^{22} \quad (17)$$

$$t_{p,e}^k = t_{lp^{12},e}^k + v_{lp^{12},e}^k \quad \forall e, p \in P11, k > 1, np^{11} + p \leq np^{11} + fp^{11} \quad (18)$$

$$t_{p,e}^k = t_{lp^{22},e}^k + v_{lp^{22},e}^k \quad \forall e, p \in P11, k > 1, np^{11} + p \leq np^{11} + fp^{11} \quad (19)$$

$$t_{p,e}^k = t_{lp^{12},e}^k + v_{lp^{12},e}^k \quad \forall e, p \in P21, k > 1, np^{21} + p \leq np^{21} + fp^{21} \quad (20)$$

$$t_{p,e}^k = t_{lp^{22},e}^k + v_{lp^{22},e}^k \quad \forall e, p \in P21, k > 1, np^{21} + p \leq np^{11} + fp^{21} \quad (21)$$

Phase Duration and Interval Constraints:

$$t_{p,e}^1 = 0 \quad \forall e, p \in P, p < SP1 \text{ or } 4 < p < SP2 \quad (22)$$

$$t_{p,e}^1 = \text{int1} \quad \forall e, p \in P, p = SP1, \text{int1} > 0 \quad (23)$$

$$t_{p,e}^1 = \text{int2} \quad \forall e, p \in P, p = SP2, \text{int2} > 0 \quad (24)$$

$$v_p^k = g_p^k + y_p + r_p \quad \forall e, p, k \quad (25)$$

$$\text{Grn1}_p^1 = \text{elapsed1} \quad \forall p \in P, p = SP1 \quad (26)$$

$$\text{Grn2}_p^1 = \text{elapsed2} \quad \forall p \in P, p = SP2 \quad (27)$$

$$1 \quad g_{p,e}^k \leq (gmax_Ring_p^k - Grn1_p^k - Grn2_p^k)I_p^k \quad \forall e, k, p \quad (28)$$

$$2 \quad g_{p,e}^k \geq (gmin_p^k - Grn1_p^k - Grn2_p^k)I_p^k \quad \forall e, k, p \quad (29)$$

3 Priority Request Constraints:

$$4 \quad d_p^j \geq t_{p,e}^1 I_p^1 + t_{p,e}^2 (1 - I_p^1) - Rl_p^j \quad \forall e, j, p, a_p^j > 0 \quad (30)$$

$$5 \quad M\theta_p^j \geq Ru_p^j - (t_{p,e}^1 + g_{p,e}^1)I_p^1 + (t_{p,e}^2 + g_{p,e}^2)(1 - I_p^1) \quad \forall e, j, p, a_p^j > 0 \quad (31)$$

$$6 \quad g_{p,e}^1 I_p^1 + g_{p,e}^2 (1 - I_p^1) \geq (Ru_p^j - Rl_p^j)(1 - \theta_p^j) \quad \forall e, j, p, a_p^j > 0 \quad (32)$$

$$7 \quad d_p^j \geq \chi_p^j - Rl_p^j \theta_p^j \quad \forall e, j, p, a_p^j > 0 \quad (33)$$

$$8 \quad \chi_p^j \leq M\theta_p^j \quad \forall e, j, p, a_p^j > 0 \quad (34)$$

$$9 \quad t_{p,e}^2 I_p^1 + t_{p,e}^3 (1 - I_p^1) - M(1 - \theta_p^j) \leq \chi_p^j \quad \forall e, j, p, a_p^j > 0 \quad (35)$$

$$10 \quad t_{p,e}^2 I_p^1 + t_{p,e}^3 (1 - I_p^1) + M(1 - \theta_p^j) \geq \chi_p^j \quad \forall e, j, p, a_p^j > 0 \quad (36)$$

$$11 \quad g_{p,e}^2 I_p^1 + g_{p,e}^3 (1 - I_p^1) \geq (Ru_p^j - Rl_p^j) \theta_p^j \quad \forall e, j, p, a_p^j > 0 \quad (37)$$

$$12 \quad Ru_p^j \theta_p^j \leq (t_{p,e}^2 + g_{p,e}^2)I_p^1 + (t_{p,e}^3 + g_{p,e}^3)(1 - I_p^1) \quad \forall e, j, p, a_p^j > 0 \quad (38)$$

13 Dilemma Zone Request Constraints:

$$14 \quad \gamma_p^j \geq t_{p,e}^1 I_p^1 + t_{p,e}^2 (1 - I_p^1) - Dl_p^j \quad \forall e, j, p, b_p^z > 0 \quad (39)$$

$$15 \quad M\theta_p^j \geq Du_p^j - (t_{p,e}^1 + g_{p,e}^1)I_p^1 + (t_{p,e}^2 + g_{p,e}^2)(1 - I_p^1) \quad \forall e, j, p, b_p^z > 0 \quad (40)$$

$$16 \quad g_{p,e}^1 I_p^1 + g_{p,e}^2 (1 - I_p^1) \geq (Du_p^j - Dl_p^j)(1 - \theta_p^j) \quad \forall e, j, p, b_p^z > 0 \quad (41)$$

$$17 \quad \gamma_p^j \geq \beta_p^j - Dl_p^j \theta_p^j \quad \forall e, j, p, b_p^z > 0 \quad (42)$$

$$18 \quad \beta_p^j \leq M\theta_p^j \quad \forall e, j, p, b_p^z > 0 \quad (43)$$

19 Flexibility Constraint:

$$20 \quad Flex = \sum_{p \in P, k \in K} (t_{p,1}^k - t_{p,2}^k) I_p^k \quad (44)$$

21 Precedence constraints (constraints 2-21) represent the precedence graph of a flexible dual-
 22 ring, eight-phase controller. Constraints 2-5 and 10-13 ensure transition of phases in the same
 23 barrier group. Barrier constraints (constraints 5-9 and 14-17) ensure phases in ring-barrier group
 24 1 are terminated at the same time. Barrier constraints 18-21 connect the phases of different cycles.
 25 These constraints ensure phases in ring-barrier group 2 terminate simultaneously so that next cycle
 26 can begin without violating traffic rules.

27 Traffic demand is dynamic in nature. The priority request can be received at any time
 28 during a cycle and may need to be updated due to the unpredictable traffic conditions. The
 29 optimization model has to be robust enough to formulate the problem at any time during any traffic
 30 conditions. Phase duration and interval constraints (constraints 22-29) are responsible for the
 31 different starting conditions. If the starting phase is in the yellow change or red clearance interval,
 32 constraint 23 and 24 represent the time required time to start the next phase green. If the starting
 33 phase is in green state, constraints 26 and 27 capture the elapsed green time data. Constraint 29
 34 ensures a phase is served for minimum amount of time before termination. Constraint 28 ensures
 35 a phase is not served more than its maximum allocated green time. Both constraints 28 and 29 are
 36 adjusted on the basis of elapsed green time. Constraint 25 ensures that there is not a shortening or
 37 omitting of the yellow change interval or red clearance intervals.

38 Priority request constraints (constraints 30-38) formulate the conditions to serve the
 39 priority request of emergency vehicles. Due to the unpredictable driving characteristics, the

vehicles expected arrival time to the intersection is considered to be in a range (earliest arrival time, R_l and latest arrival time, R_u). This range-based expected arrival time increases the robustness of the model. Constraint 30 calculates the delay of the emergency vehicles priority request. During the cycle in which the priority request is served, the ending green time ($t_{p,e}^k + g_{p,e}^k$) of the requested phase cannot be less than the expected arrival time of emergency vehicle. Constraint 31 addresses this requirement. A binary variable θ_p^j is used to address the combinatorial optimization problem. Constraints 32, 33, 37 and 38 are relaxed due to the binary variable. These constraints determine the cycle during which the request will be served by the associated service phase. If priority request is not served in the first cycle, constraint 32 ensures the allocated green time of the requested phase in the first cycle is at least equal to the time interval between the latest arrival time and the earliest arrival time. An intermediate variable ($\chi_p^j = \theta_p^j t_{p,e}^k$) is introduced in constraints 33-36 to linearize the model.

Dilemma zone request constraints 39-43 ensure that the request is served in the first cycle. This is because a dilemma zone only exists if the phase is green for the freight vehicle. These constraints are similar to the priority request constraints 30-34. They ensure the safe passage of the dilemma zone vehicles.

Flexibility constraint 44 allows for the implementation of a flexible actuated signal timing. This constraint creates the left and right critical points in the signal schedule. The feasible region between the left and right critical points is the time period when actuated signal logic can be implemented. The flexible signal control strategy reduces the negative impact on the regular vehicles.

Analysis of Schedule

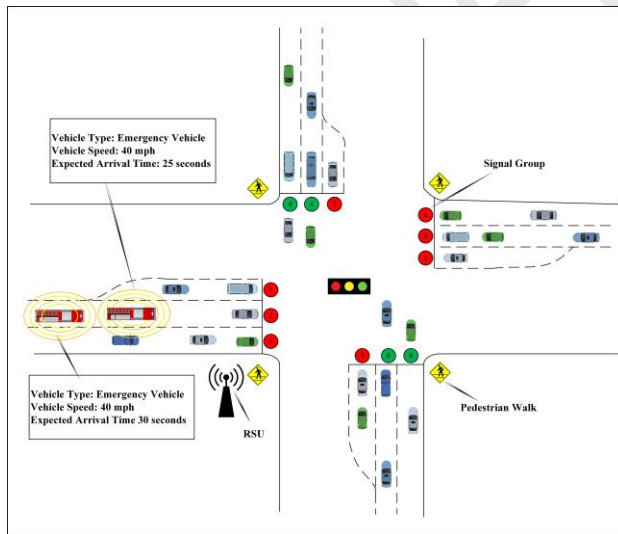
The optimization model provides a traffic signal timing schedule based on the current situation. To analyze the model and optimal signal control strategy, several specific situations are considered. Figure 4 depicts four scenarios where emergency vehicles are approaching from same or different directions. For each situation, a time-phase diagram illustrates the flexible optimal signal control strategy obtained from the proposed preemption algorithm and optimization model. In each time-phase diagram, the vertical left and right axis denote signal phases in ring 1 and ring 2, respectively, and the horizontal axis denotes the phase duration (time). The optimal signal schedule has left critical points (CP_l) and right critical points (CP_r) for the phases in both rings. A phase is to be until the left critical points (determined by phase's minimum green time or by the optimal schedule) and the phase will be terminated at the right critical point (determined by phase's maximum green time or when the phase needs to be terminated so that subsequent phases can be served before the priority request is served). The region between the left and right critical points denote the flexible region where phases can be extended based on the traffic demand.

In Figure 4(a), multiple (two) connected emergency vehicles are approaching from the eastbound approach. When the PRS receives the priority requests from the connected emergency vehicles, the through phases (phase 4, and 8) for northbound and southbound approaches are green. The red rectangles in Figure 4(b) denote when the emergency vehicles will reach the intersection at $t=25$ seconds and $t=29$ seconds respectively (priority requests are displayed for two cycles since the model determines if they are served in the first or second cycle, but their ETA is the same regardless of when they are served). The time-phase diagram (Figure 4(b)) shows that phase 4 and phase 8 can be terminated any time between 5 and 21 seconds. This flexible signal timing plan will reduce the overall intersection's vehicle delay and ensure minimal priority delay of the

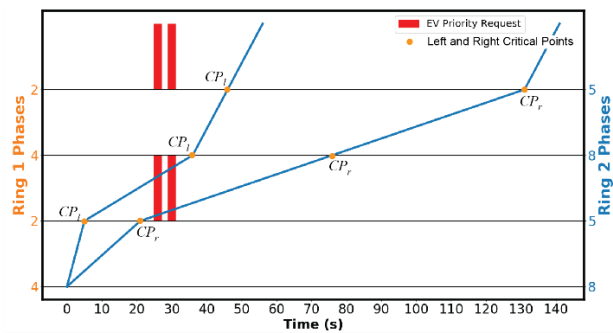
connected emergency vehicles. (Note: the ring 1 and ring 2 phase durations are overlaid in the figure).

Figure 4(c) depicts a scenario where a connected emergency vehicle and a connected truck are approaching from different approaches (eastbound approach and southbound approach) at a speed 40 mph and 35mph, respectively. Both vehicles are sending priority request to the signal controller. The green and red rectangles in Figure 4(d) indicate the dilemma zone request of truck and priority request of emergency vehicle for two cycles, respectively. The truck is approximately 210 feet away from the intersection and signal phase (phase 4) status is green. The emergency vehicle is approximately 600 feet away from the intersection and the desired service phase signal status (phase 2 and 5) is red. Due to physical and deceleration characteristics of the truck it is unsafe to force-off the southbound signal phase (phase 4). Figure 4(d) denotes that the optimal signal timing is to hold the signal phase green for the truck so that it can pass the intersection safely. The signal phase will force-off as soon as the truck passes the intersection. This signal timing causes 1 second delay for the emergency vehicle but it provides some assurance of traffic safety by not terminating the green for the dilemma zone truck.

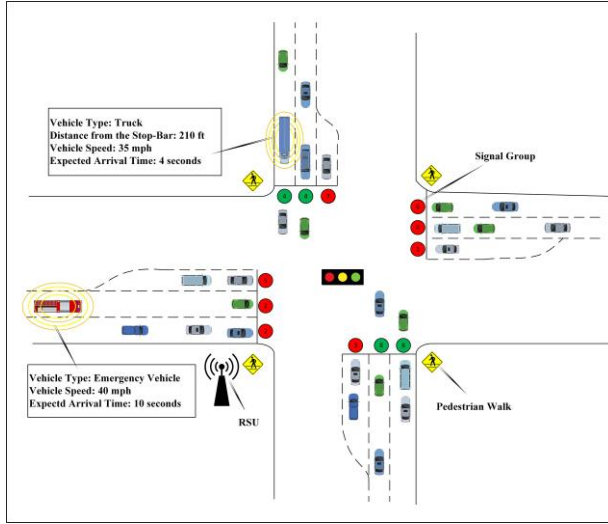
Figure 4(e) and 4(g) illustrate almost the same scenario where an emergency vehicle is approaching from the eastbound approach and another emergency vehicle and a truck trailer are approaching from the southbound approach. In Figure 4(g), the truck is in the dilemma zone whereas in Figure 4(e), it is approximately 450 feet away from the intersection. The time-phase diagram of the first scenario (Figure 4(e)) indicates that the truck is not in the dilemma zone. To minimize the priority delay, it is better to force-off the signal phases for the southbound approach and serve the emergency vehicle on the eastbound approach first. After serving the emergency vehicle on the eastbound approach, the signal controller will serve the emergency vehicle on the southbound approach. But in case of the second scenario (Figure 4(g)), the truck is in dilemma zone. The time-phase diagram (Figure 4(h)) indicates that optimal signal timing is to serve the emergency vehicle on the southbound approach along with the truck first.



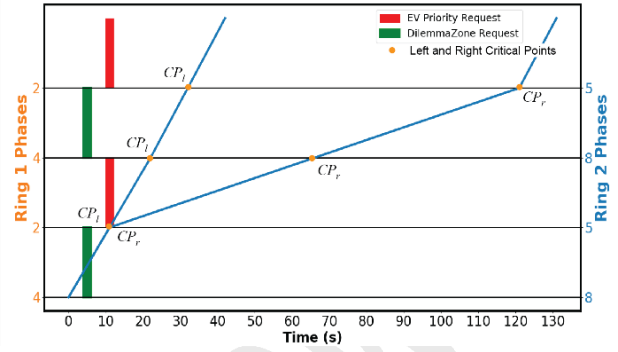
(a)



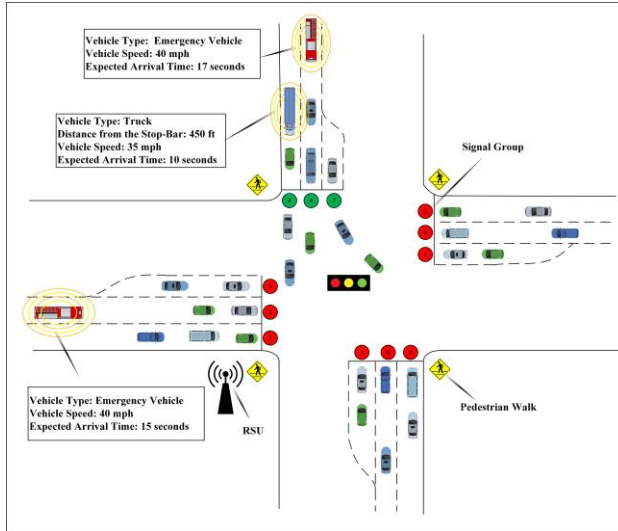
(b)



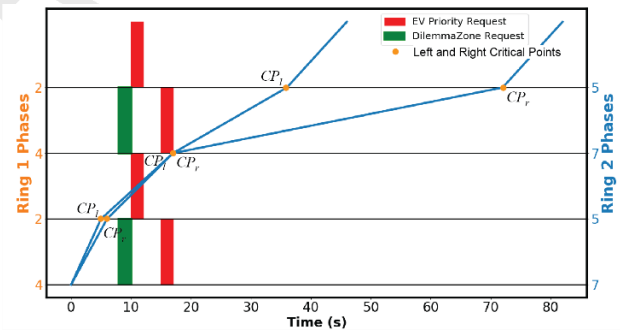
(c)



(d)



(e)



(f)

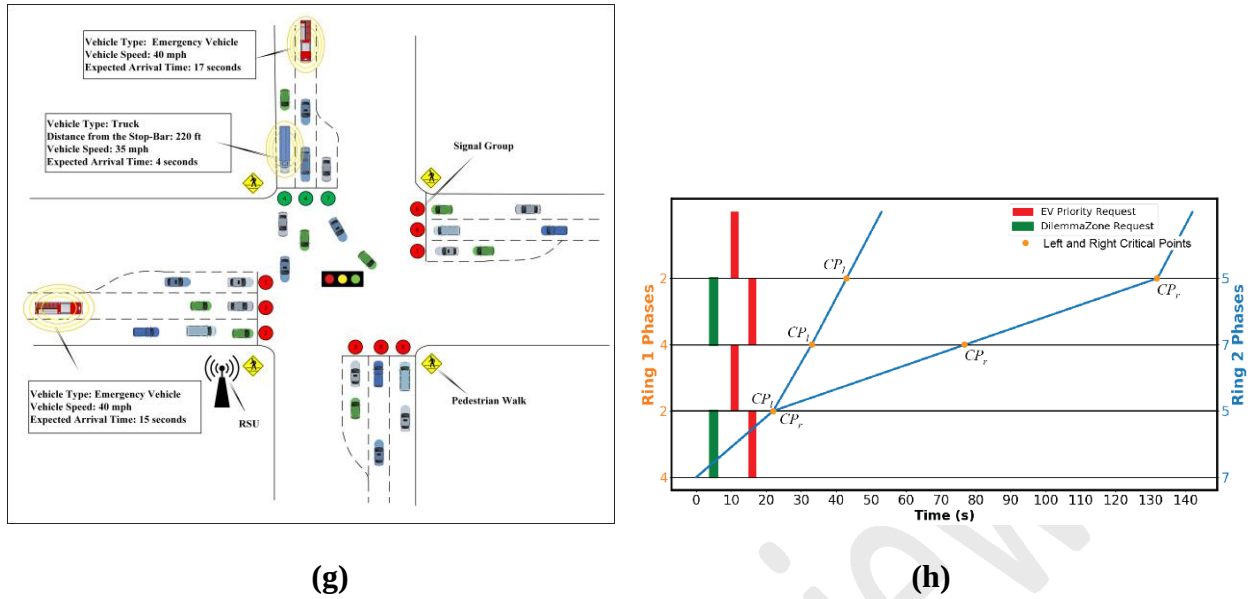


Figure 4: (a) Two Emergency vehicles are approaching from the eastbound approach, (c) An Emergency Vehicle and a Truck are approaching from different directions (e), (g) An Emergency vehicle is approaching from the eastbound approach and an Emergency vehicle and a Truck are approaching from the southbound approach, (b), (d), (f), and (h) Time-Phase Diagram of (a), (c), (e), and (g) scenarios respectively.

NUMERICAL EXPERIMENTS

Numerical experiments were conducted to evaluate the proposed emergency vehicle priority control strategy. A microscopic simulation, VISSIM is used to compare the performance of the different algorithm. To simulate real world traffic signal controller, the Econolite ASC/3 software in the loop controller is installed with VISSIM. A VISSIM model based on a test network in Anthem, AZ was created and calibrated to replicate an emergency incident (Figure 5). The emergency incident occurs just the west of Daisy- Gavilan intersection. Four emergency vehicles (EV1, EV2, EV3 and EV4) are initially located at different origins on the network. EV1 and EV2 start from the Daisy-Memorial fire station (FS1) and travel through five intersections to reach the response destination. EV3 and EV4 are initially located east of the Daisy-Meridian (FS2) and travel through three intersections to reach the response destination. The actual signal timing data is provided by Maricopa County Department of Transportation (MCDOT). Traffic demand is relatively low in Anthem. The proposed algorithm was tested in under-saturated traffic condition. A GPS based emergency vehicle preemption (EVP) system was modeled for the traditional preemption system. This system allows the elimination of the line-of-sight requirements associated with the optical-based system (43). A software component, Message Distributor and a custom driver model (develop by using the original sample driver-model provided with VISSIM distribution) are utilized to simulate connected vehicle environment. The emergency vehicles reach the destination in approximately 3-3.5 minutes. The simulation runs for 1.5-2 minutes beyond their arrival to capture the impact of preemption on the signalized intersections, but not for longer so that the transient behavior is captured. The warm-up time for simulation model is 20 minutes. Ten simulation replications are used to estimate the average travel time, average vehicle delay, and average stopped delay to evaluate the performance of the proposed algorithm compare to the GPS preemption system.

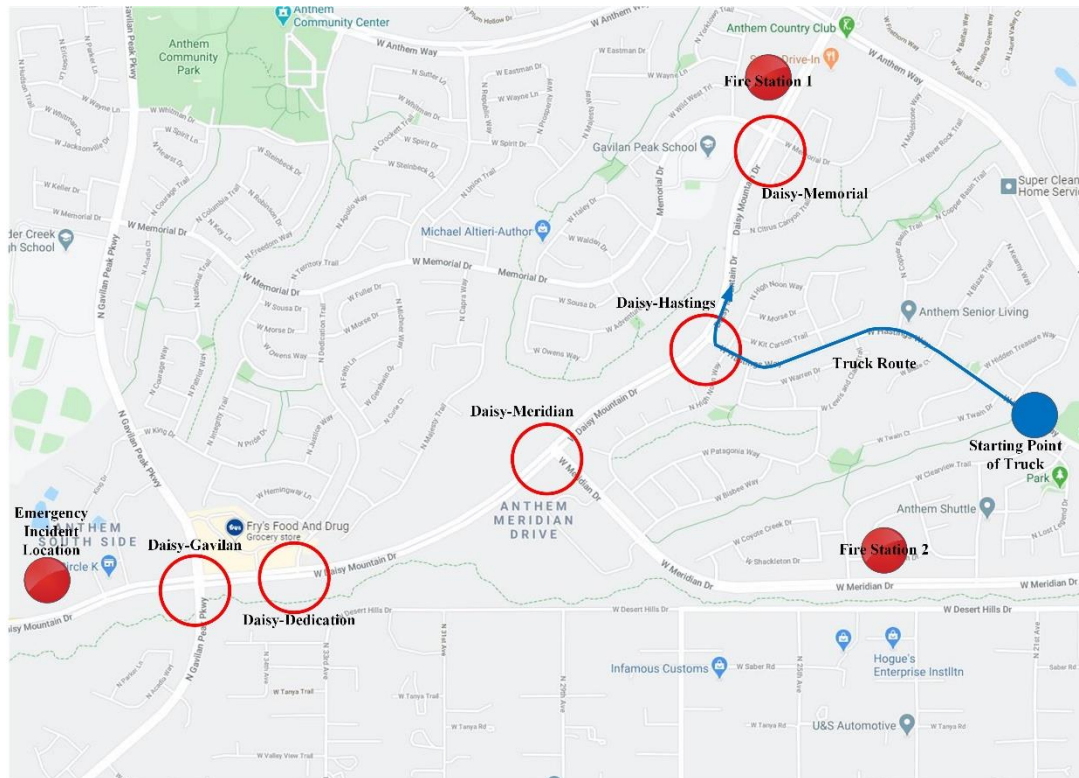
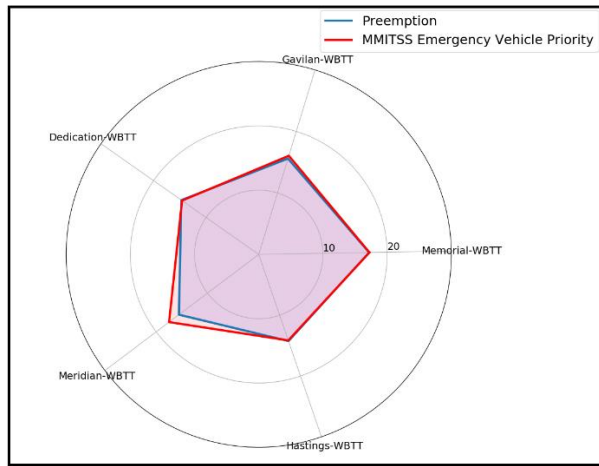


Figure 5: Overview of Anthem networks (Source: Google Maps)

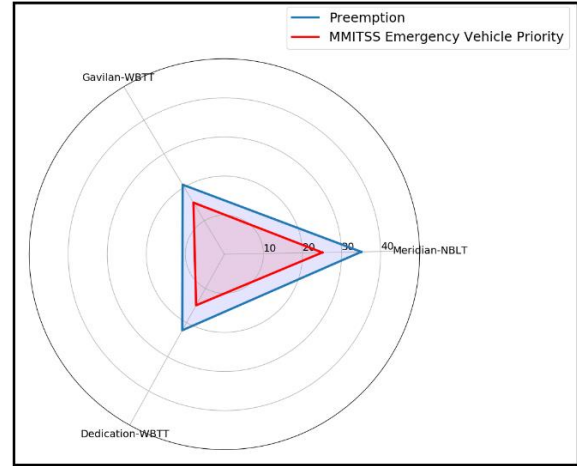
Radar diagram are utilized to visualize the average travel time comparison between GPS preemption and the MMITSS priority algorithm. On each axis of the diagram, the movements at an intersection are shown. Figure 6(a) and 6(b) depict the travel time comparison between the emergency vehicles. For emergency vehicles from FS1, the MMITSS priority performs as well as GPS preemption system, except for a slight increase in travel time at Meridian. For vehicles from FS2, MMITSS priority performs significantly better. EV1, EV2 and EV3, EV4 approach Daisy-Meridian intersection simultaneously. But in each simulation run, their arrival time are different due to the stochastic nature of VISSIM. In the signal controller, preemptor 1 is the highest priority preemptor (44). Preemptor 1 is assigned to the westbound approach (major-street) and preemptor 2 in the northbound approach (minor-street) of the Daisy-Meridian intersection. In the GPS preemption system, EV1 and EV2 always gets priority due to the priority hierarchy. The MMITSS priority algorithm considers vehicles expected arrival time (ETA) at the intersection when multiple EVs approach simultaneously. In some simulation run, MMITSS priority allows EV3 and EV4 to pass the intersection before EV1 and EV2. Due to this, the average travel time of EV1 and EV2 are slightly higher at the Daisy-Meridian intersection for MMITSS.

The impact of both algorithms on regular vehicles is analyzed in Figure 6(c)-6(g) per intersection. The average vehicle travel times are sometimes better for GPS preemption on the westbound through movement (WBTT). GPS preemption provides green indication immediately after receiving the priority requests which reduces the travel time of the regular vehicle too. MMITSS priority provides green indication to the EV routes based on optimal signal schedule. If compatible through phases along the EV route are green, MMITSS does not terminate the phases to provide green indication to the through and left turn phases for the EV route. This strategy benefits the vehicles travel along the eastbound through movement (EBTT).

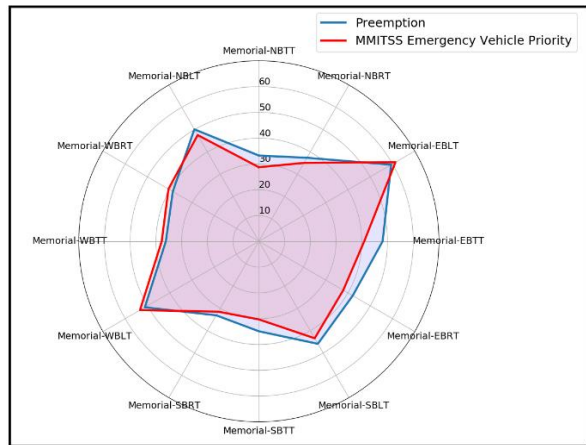
Figure 6(h) illustrate the benefits of the proposed algorithm on the minor-street for all intersections. The optimal signal schedule of MMITSS priority reduces the negative impact on the minor-street without delaying the emergency vehicles.



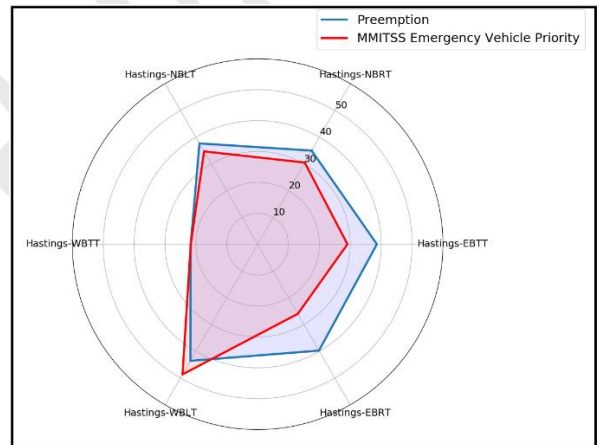
(a)



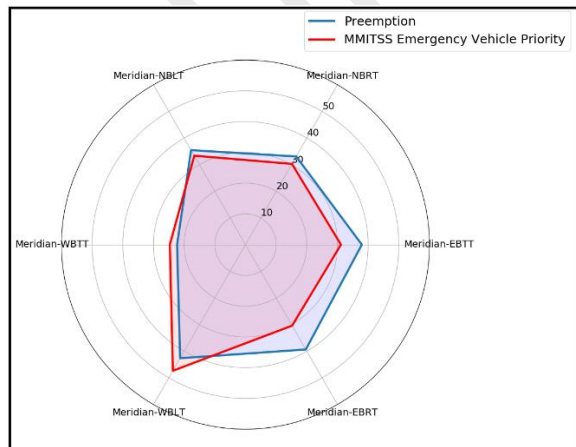
(b)



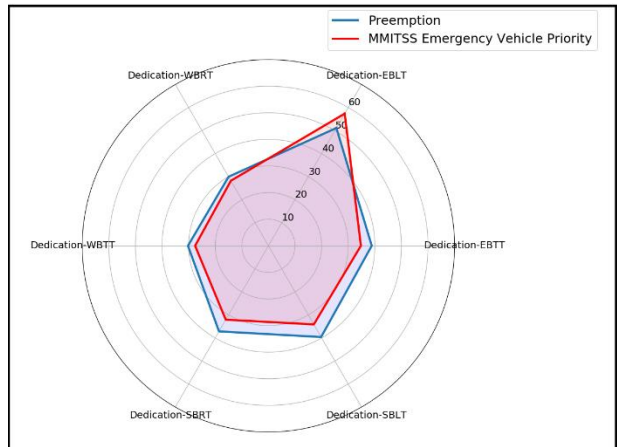
(c)



(d)



(e)



(f)



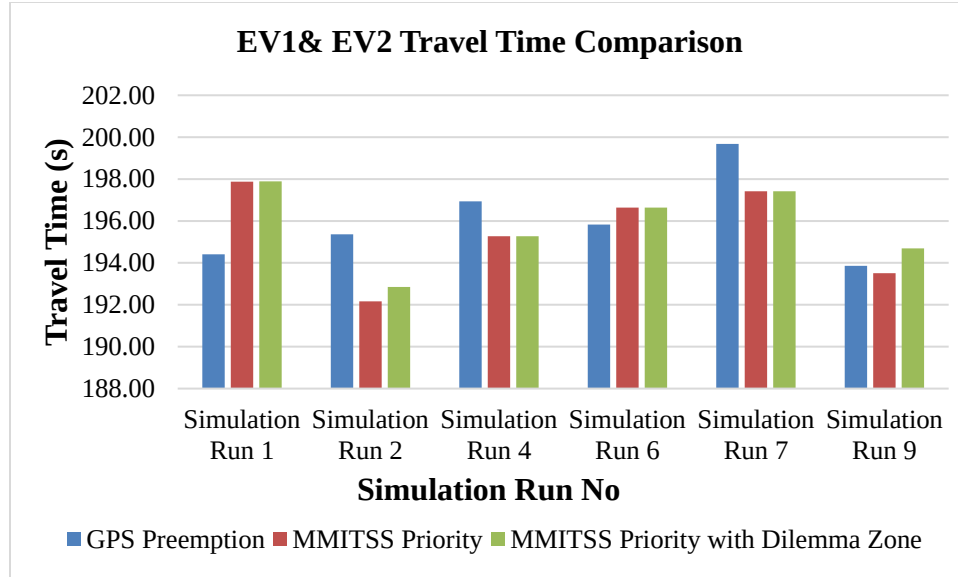
Figure 6: Comparison of Travel Time between GPS Preemption and MMITSS Priority for (a) EV1 and EV2 (b) EV3 and EV4 (c) Daisy-Memorial Intersection (d) Daisy-Hastings Intersection (e) Daisy-Meridian Intersection (f) Daisy-Dedication Intersection (g) Daisy-Gavilan Intersection (h) Minor-Street

Table 2 summarizes the benefits of MMITSS priority compared with GPS preemption on the overall intersection performance. The average vehicle delay and average stopped delay are decreased by at least 10% for all intersections.

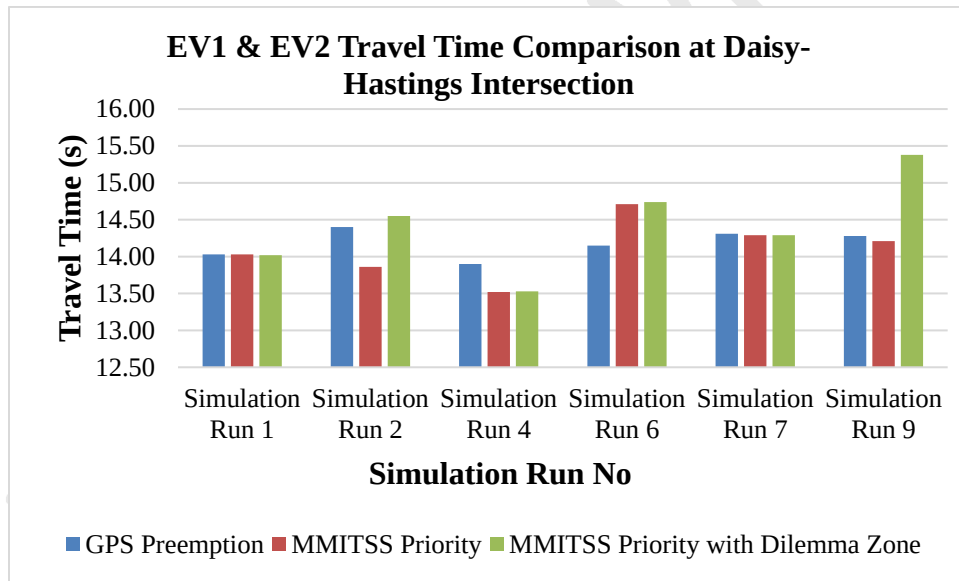
Table 2: Average vehicle delay and average stopped delay comparison

Intersection Name	GPS Preemption		MMITSS Priority		% Of Change	
	Average Vehicle Delay (s)	Average Stopped Delay(s)	Average Vehicle Delay (s)	Average Stopped Delay(s)	Average Vehicle Delay (s)	Average Stopped Delay(s)
Daisy-Memorial	27.25	18.92	24.44	16.40	-10.32	-13.31
Daisy-Hastings	22.30	14.92	15.86	10.61	-28.90	-28.89
Daisy-Meridian	18.23	11.86	15.33	9.36	-15.91	-21.05
Daisy-Dedication	21.16	14.38	19.02	12.73	-10.12	-11.47
Daisy-Gavilan	30.64	23.10	26.98	20.58	-11.94	-10.90

The MMITSS priority algorithm can consider the presence of connected heavy vehicles in the dilemma zone as a dilemma zone priority request. In six of the ten simulation runs, EV1 and EV2 send priority requests when a connected truck is in the dilemma zone on the northbound approach at the Daisy-Hastings intersection. It is possible to specify the weight of the different priority requests in the model. If the weight of the dilemma zone priority request is higher, the optimal signal will allow the truck to pass the intersection before EV1 and EV2. EV1 and EV2 travel time comparison between GPS Preemption, MMITSS Priority and MMITSS Priority with Dilemma Zone Algorithm for the whole route and at Daisy-Hastings intersection is shown in Figure 7.



(a)



(b)

Figure 7: EV1 & EV2 Travel Time comparison for different priority weight and algorithm: (a) whole EV Route (b) Daisy-Hastings Intersection

CONCLUSION

This paper presents an emergency vehicle priority control systems which utilizes connected vehicle technologies. This system has a potential to reduce the negative impact on the minor-streets and conflicts between the emergency vehicles and the heavy vehicles that are in the dilemma zone. The approach considers the priority requests of the emergency vehicles, and heavy vehicles that are in the dilemma zone in a MILP model. The optimization model provides a flexible optimal solution which can be implemented by considering real-time vehicle actuation. The model has

been tested and evaluated using microscopic traffic simulation. The simulation experiments show that the proposed model can ensure the smooth progression of the emergency vehicles without having a significant negative impact on the non-emergency vehicles. Numerical experiments also show that the proposed method can decrease the average vehicle delay and average stopped delay by at least 10% for all intersections. The proposed emergency vehicle priority systems will be implemented the field during the fall of 2020.

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AUTHOR CONTRIBUTION

The authors confirm contribution to the paper as follows: study conception and design: Larry Head, Faisal Saleem, Debashis Das; system design and development: Debashis Das, Niraj Altekar; testing: Debashis Das, Larry Head, Niraj Altekar; draft manuscript: Debashis Das, Larry Head. All authors reviewed the results and approved the final version of the manuscript. The authors do not have any conflicts of interest to declare.

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